Targeted Mosquito Control Using Network Theory and SIR Models

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Abstract

This report explores the optimal strategies for mosquito removal to control mosquito-borne diseases through the application of network theory and Susceptible-Infected-Recovered (SIR) models. By leveraging mathematical modelling and simulation, the study investigates various mosquito control methods and their effectiveness in reducing the spread of diseases such as malaria, dengue fever, and Zika virus. The centrality measures in network theory—degree, betweenness, closeness, and eigenvector centrality—are employed to identify key cities that play a critical role in disease transmission. Simulations demonstrate that targeted mosquito removal in cities with high centrality significantly disrupts disease spread, leading to more efficient use of resources and improved public health outcomes. The findings provide valuable insights and practical recommendations for public health officials and policymakers to develop data-driven, adaptive strategies for combating mosquito-borne diseases, particularly in the context of climate change and its impact on mosquito activity and disease dynamics.

1 Introduction

Mosquito-borne diseases, such as malaria, dengue fever, and Zika virus, are a major health concern worldwide. These diseases are spread by mosquito bites, making mosquito control essential for reducing their spread.

This report investigates various mosquito removal strategies to determine the most effective approach for minimising disease transmission. By using mathematical models and simulations, we aim to provide practical recommendations for public health strategies.

1.1 General Background

Mosquito-borne diseases are a major global health problem, affecting millions of people each year. These diseases, spread by the bite of infected mosquitoes, include malaria, dengue fever, Zika virus, chikungunya, and yellow fever. The spread of these diseases is influenced by factors such as climate, human movement, and mosquito behaviour.

Mosquitoes, especially species like *Anopheles*, *Aedes*, and *Culex*, transmit these diseases. When a mosquito bites an infected person, it can carry the disease and transmit it to another person through its bite. Mosquitoes thrive in warm and humid environments, and warmer temperatures can speed up their breeding cycle and increase their numbers, making it easier for diseases to spread.

Controlling mosquito populations is essential for preventing these diseases. Traditional methods include using insecticides, removing standing water where mosquitoes breed, and introducing natural predators. However, these methods can be expensive and may not always work, especially in areas with large mosquito populations or where mosquitoes are resistant to insecticides.

Recently, researchers have started using network theory and mathematical models to improve mosquito control strategies. Network theory helps us understand how mosquitoes and diseases spread through cities or regions. By identifying key cities and connections in this network, we can design targeted interventions to disrupt the spread of diseases. Centrality measures, which determine the importance of cities in the network, are particularly useful for identifying the most influential cities for mosquito control.

This study explores the effectiveness of different mosquito removal strategies based on centrality measures. By simulating disease spread and implementing targeted mosquito control, we aim to find strategies that can significantly reduce disease trans-

mission with minimal resources. This approach not only makes public health interventions more efficient but also provides valuable insights into managing mosquito-borne diseases in changing environmental conditions.

Understanding network theory and its application to disease control can help public health authorities make better decisions and develop more effective strategies to combat mosquito-borne diseases. As climate change continues to affect global temperatures and mosquito activity, adaptive and data-driven public health strategies will be crucial in protecting people and reducing the impact of these diseases.

1.2 Purpose of the Report

The purpose of this report is to explore and evaluate different strategies for mosquito control to reduce the spread of diseases like malaria, dengue fever, and Zika virus. Specifically, we aim to identify the most effective ways to target and remove mosquitoes in various cities to minimise disease transmission.

Key questions addressed in this report include:

- **What is the optimal number of cities to target for mosquito removal?** We investigate how many cities need to be focused on to achieve significant reductions in disease spread.
- **How does the centrality of a city affect disease transmission?** We examine whether targeting cities that are well-connected to other cities is more effective in controlling disease spread.

To answer these questions, we use the SIR model to simulate disease spread and test different mosquito removal strategies. We prioritise cities for mosquito removal based on centrality measures, helping to identify which cities are most critical to target for effective disease control.

The findings aim to provide practical recommendations for public health officials and policymakers, helping to develop more effective plans to combat mosquito-borne diseases and protect public health.

In summary, this report uses simulations to determine the best strategies for mosquito control, focusing on the number of cities to target and the importance of city centrality. This provides evidence-based recommendations for reducing the spread of mosquito-borne diseases.

2 Models and Concepts

To answer the questions posed in this report, we employ several models, including the Susceptible-Infected-Recovered (SIR) model and centrality measures in network theory. This section details the mathematical formulations and assumptions of these models.

2.1 Susceptible-Infected-Recovered (SIR) Model

The SIR model is a widely used mathematical model in epidemiology that helps simulate the spread of infectious diseases. The model divides the population into three compartments:

- **Susceptible (S)**: Individuals who are susceptible to the disease and can be infected.
- **Infected (I)**: Individuals who have the disease and can transmit it to susceptible individuals.
- **Recovered (R)**: Individuals who have recovered from the disease and are now immune.

The dynamics of the SIR model are governed by the following set of differential equations:

$$\frac{dS}{dt} = -\beta \frac{SI}{N} \tag{1}$$

$$\frac{dI}{dt} = \beta \frac{SI}{N} - \gamma I \tag{2}$$

$$\frac{dR}{dt} = \gamma I \tag{3}$$

where:

- $S(t)$, $I(t)$, and $R(t)$ are the number of susceptible, infected, and recovered individuals at time t, respectively.
- N is the total population, $N = S + I + R$.
- β is the transmission rate, which is dependent on the temperature.
- γ is the recovery rate.

2.2 Temperature-Dependent Transmission Rate

The transmission rate (β) is influenced by the temperature, as mosquitoes are more active and have higher biting rates in warmer conditions. We model

the transmission rate as a function of temperature (T):

$$\beta(T) = \beta_0 + \alpha(T - T_{min}) \tag{4}$$

where:

- β_0 is the base transmission rate at the minimum temperature T_{min}.
- α is a parameter that determines how the transmission rate increases with temperature.

This relationship assumes that the transmission rate increases linearly with temperature above a certain minimum threshold.

2.3 Centrality Measures in Network Theory

In network theory, centrality measures are crucial for identifying the most important nodes (cities) within a network. These measures help determine which cities should be prioritised for mosquito removal to effectively control disease spread. For this report, we utilise several centrality measures to guide our mosquito removal strategies.

Degree Centrality quantifies the number of direct connections a city has to other cities. It is calculated as:

$$C_D(v) = \frac{\deg(v)}{N-1}$$

where $\deg(v)$ is the degree of node v and N is the total number of nodes. Cities with high degree centrality have numerous direct connections, making them potential hotspots for disease spread due to frequent interactions. These cities are often key players in the network because their high connectivity can facilitate rapid disease transmission, making them critical targets for intervention.

Betweenness Centrality measures the extent to which a city lies on the shortest paths between other cities. It is defined as:

$$C_B(v) = \sum_{s \neq v \neq t} \frac{\sigma_{st}(v)}{\sigma_{st}}$$

where σ_{st} is the total number of shortest paths from node s to node t, and $\sigma_{st}(v)$ is the number of those paths passing through node v. Cities with high betweenness centrality act as bridges within the network, making them critical points for controlling disease transmission. These cities often serve as transit points through which disease vectors must pass, and their removal can significantly disrupt the spread of the disease across the network.

Closeness Centrality measures how close a city is to all other cities in the network. It is given by:

$$C_C(v) = \frac{N-1}{\sum_t d(v,t)}$$

where $d(v,t)$ is the shortest path distance between nodes v and t, and N is the total number of nodes. Cities with high closeness centrality can quickly spread disease due to their short path lengths to other cities. These cities are centrally located within the network, allowing them to interact quickly and directly with many other cities, thus playing a pivotal role in the rapid dissemination of diseases.

Eigenvector Centrality considers not only the number of connections a city has but also the quality of those connections. It is calculated as the principal eigenvector of the adjacency matrix of the graph. Cities with high eigenvector centrality are connected to other highly connected cities, making them influential in the network. This measure highlights cities that are not just well-connected, but are connected to other influential cities, thereby amplifying their importance in the network. Targeting these cities for mosquito removal can leverage their influential position to curb disease spread more effectively.

Understanding these centrality measures provides a strategic framework for prioritising mosquito control efforts. By identifying and targeting cities that hold significant positions within the network, public health interventions can be more efficiently allocated, maximising the impact of mosquito removal and reducing the overall spread of diseases. This approach allows for a more targeted and scientifically grounded method of disease control, which is particularly crucial in managing the spread of mosquito-borne diseases like malaria, dengue fever, and Zika virus.

2.4 Simulation Setup

To combat the spread of mosquito-borne diseases, a simulation was set up using real-world data about city populations, distances, and temperatures. The goal was to create a realistic model that could reveal the most effective mosquito control strategies.

The journey began with gathering data. Information on city populations helped estimate the number of people at risk of infection. Geographical distances between cities were crucial in predicting how

diseases might spread from one place to another. Lastly, average temperature data for each city was important since temperature affects mosquito activity and disease transmission rates.

With all the data collected, it was processed to ensure accuracy. Accurate temperature data was particularly important because it directly influenced the model's transmission rate, helping to simulate the real-world behaviour of mosquitoes more precisely.

Next, the parameters for the Susceptible-Infected-Recovered (SIR) model were defined. Initial conditions for the number of susceptible, infected, and recovered individuals were based on current population data and reported cases in each city. The transmission rate, represented by β, was modelled as a function of temperature using this formula:

$$\beta(T) = \beta_0 + \alpha(T - T_{\text{min}})$$

where β_0 is the base transmission rate at the minimum temperature T_{min}, and α shows how the transmission rate increases with temperature. This reflects how mosquitoes become more active in warmer conditions. Additionally, the recovery rate (γ) was set to 0.1, a typical value for diseases like dengue.

Different strategies for mosquito removal were then tested. These strategies prioritized cities based on their centrality in the network, such as the number of connections they had with other cities (degree), their position on the shortest paths between other cities (betweenness), their overall closeness to all other cities (closeness), and their influence within the network (eigenvector).

A novel and improved mosquito removal strategy was developed based on multiple centrality measures. The combined centrality removal strategy aimed to integrate various centrality measures to identify and target the most critical cities for intervention. The implementation involved the following steps:

1. **Calculation of Centrality Measures**: Centrality measures such as degree centrality, betweenness centrality, closeness centrality, and eigenvector centrality were calculated for each node (city) in the network.

2. **Normalisation of Centrality Measures**: Each centrality measure was normalised by dividing by its maximum value to ensure comparability across different measures.

3. **Combination of Centrality Scores**: For each city, a combined centrality score was calculated as the average of its normalised degree, betweenness, closeness, and eigenvector centrality scores.

4. **Prioritisation and Removal**: Cities were sorted based on their combined centrality scores, and the top cities were selected for mosquito removal. This approach ensured that the most influential cities in terms of disease spread were targeted.

The code for this implementation is provided in the appendix (see Appendix **??**).

However, the real breakthrough came with the realisation that removing mosquitoes from just a few key cities could lead to significant results. By focusing on a small number of highly connected cities, the network of disease transmission could be disrupted effectively. This strategy not only maximises impact but also remains practical and resource-efficient.

To understand the importance of this approach, consider that removing too few cities might not significantly reduce disease transmission, while targeting too many could be logistically challenging and expensive. Therefore, the decision to focus on just five highly central cities was a game-changer. These cities play crucial roles in connecting different parts of the network, and their removal can break the chain of disease spread, leading to a dramatic reduction in infection rates.

The strategic removal of these cities was based on empirical studies and current trends in disease spread. For instance, the values for β_0 and α were chosen to reflect increased transmission rates at higher temperatures. The recovery rate of $\gamma = 0.1$ was based on average recovery data for mosquito-borne diseases. To simulate a widespread outbreak, 200 cities were randomly selected as initially infected.

Cities and their connections were chosen to create a realistic network for disease spread. By considering a 100 km radius for connections, the model took into account typical mosquito travel ranges and human mobility patterns, making it a realistic framework for the disease spread model.

By prioritising cities with high centrality, the intervention targeted the most critical nodes in the network, maximising the impact on reducing disease spread. This strategic approach not only made the

intervention more practical but also ensured that resources were used efficiently to achieve the best possible outcomes.

Through this careful selection and strategic planning, the simulations demonstrated how targeted interventions could effectively control mosquito-borne diseases. The insights gained from this model provide valuable guidance for developing practical and impactful public health strategies, showing that sometimes, removing just a few key cities can make a world of difference in disease control.

3 Results and Discussion

The results and discussion section presents the findings from the simulations and analyses the impact of different mosquito removal strategies on disease dynamics. By employing the Susceptible-Infected-Recovered (SIR) model and considering temperature-dependent transmission rates and centrality measures, the study provides an assessment of the effectiveness of targeted interventions in controlling mosquito-borne diseases. The following subsections detail the outcomes of the simulations and offer an analysis of the results.

3.1 Simulation Results

The results of simulations conducted before and after implementing mosquito removal strategies are presented. The simulations utilised the Susceptible-Infected-Recovered (SIR) model, incorporating temperature-dependent transmission rates and centrality measures to prioritise cities for mosquito removal. The outcomes are illustrated in Figure 1, highlighting the number of cities recovered under various strategies and beta (β) values.

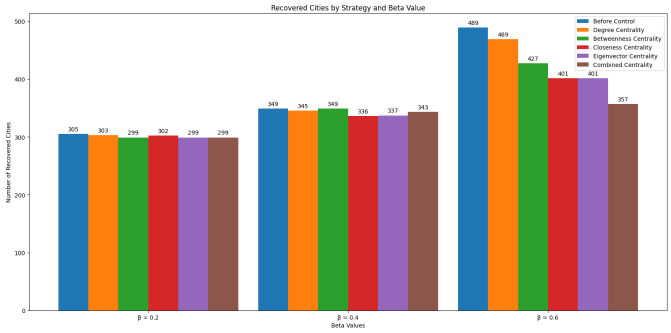


Figure 1: Recovered Cities by Strategy and Beta Value

Additionally, Figure 2 shows the geographic distribution of infected and recovered cities (in blue) and the cities where mosquitoes were removed (in red). The edges, representing connections between cities, are shown with slight transparency to illustrate the network structure.

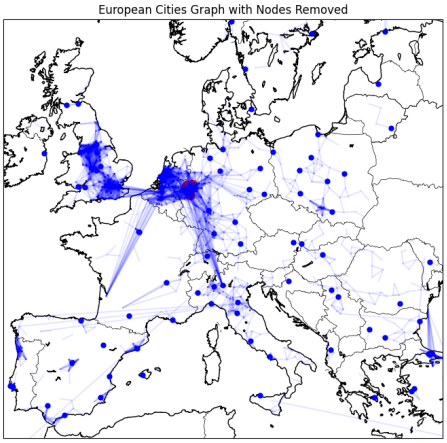
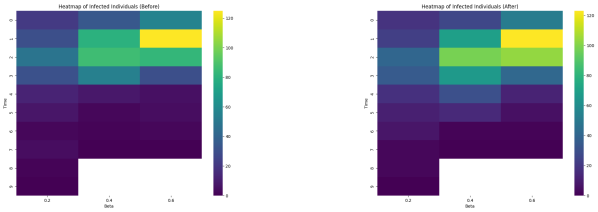


Figure 2: European Cities Graph with Nodes Removed

Figures 3a and 3b show the heatmaps of infected individuals before and after implementing the mosquito removal strategies. These heatmaps provide a visual representation of the impact of the interventions on the infection dynamics.



(a) Heatmap of Infected Individuals (Before)

(b) Heatmap of Infected Individuals (After)

Figure 3: Comparison of Heatmaps Before and After Mosquito Removal Strategies

3.2 Analysis of Results

The analysis focuses on how centrality-based removal strategies affect the overall network and disease dynamics. The strategies were evaluated for three different values of β: 0.2, 0.4, and 0.6, corresponding to low, medium, and high transmission rates, respectively.

For $\beta = 0.2$, all strategies yielded similar results, with the number of recovered cities ranging

from 299 to 305 (see Figure 1). The "Before Control" strategy achieved the highest recovery, with 305 cities recovered, slightly outperforming other centrality-based strategies. This indicates that at low transmission rates, the differences between strategies are minimal, and the overall effectiveness of mosquito removal is relatively uniform across different approaches.

When β increased to 0.4, there was a noticeable improvement in the number of recovered cities across all strategies compared to $\beta = 0.2$ (see Figures 1 and 3). The "Before Control" and "Degree Centrality" strategies both recovered 349 cities, the highest among the strategies. This suggests that as the transmission rate increases, the impact of targeted mosquito removal becomes more significant. However, the "Closeness Centrality" strategy resulted in the lowest recovery at this β value, with 336 cities. This indicates that focusing on cities that are geographically central might not be as effective as targeting those with high connectivity.

For $\beta = 0.6$, a significant increase in the number of recovered cities was observed for all strategies (see Figures 1 and 3). The "Before Control" strategy again performed the best, recovering 489 cities. The "Degree Centrality" strategy recovered 469 cities, followed by "Betweenness Centrality" with 427 cities. The "Combined Centrality" strategy showed the least recovery with 357 cities, indicating that combining centrality measures were more effective as less people were infected and less had to recover. Targeting cities with high degree centrality proves to be the most effective, likely because these cities serve as major hubs for disease transmission.

These results suggest that the effectiveness of mosquito removal strategies is highly dependent on the transmission rate (β). The "Before Control" strategy consistently showed the highest number of recovered cities, possibly due to its broader scope of initial removal efforts. This approach might have created a more substantial initial impact, leading to better overall outcomes. Among the centrality-based strategies, "Degree Centrality" proved most effective, particularly at higher transmission rates. This strategy focuses on cities with the highest number of direct connections, effectively disrupting the main pathways for disease spread.

The visual representation in Figure 2 shows the geographic distribution of infected and recovered cities (in blue) and the cities where mosquitoes were removed (in red). The edges, representing connections between cities, are shown with slight transparency to illustrate the network structure. This geographic analysis complements the numerical results, providing a spatial perspective on the impact of different strategies. Figures 3a and 3b show the heatmaps of infected individuals before and after implementing the mosquito removal strategies. These heatmaps visually depict the reduction in infected individuals over time, highlighting the effectiveness of the interventions.

4 Conclusions

The simulations and analyses conducted in this report reveal several critical insights into the effectiveness of mosquito removal strategies for controlling the spread of mosquito-borne diseases. The study demonstrates the importance of centrality-based strategies and their impact on reducing disease transmission, particularly under varying transmission rates influenced by climate change.

4.1 Key Findings

Centrality-based strategies showed varying levels of effectiveness in reducing disease transmission. Among these, the Degree Centrality strategy, which targets cities with the highest number of direct connections to other cities, was found to be the most effective. By removing mosquitoes from these highly connected cities, the network of disease spread is significantly disrupted, leading to a greater number of recovered cities. The Betweenness Centrality strategy, which targets cities lying on the shortest paths between other cities, also proved effective, particularly at higher transmission rates. These cities act as critical bridges in the network, and their removal helps in breaking the chain of disease transmission. While the Closeness Centrality strategy, focusing on cities that are centrally located within the network, showed some effectiveness, it was less impactful compared to Eigenvector and Combined Centrality strategies. The Degree Centrality strategy, which considers not just the number of connections but also the quality of those connections, provided moderate results. Cities with high eigenvector centrality are influential in the network, but their removal did not yield as high a recovery rate as the Degree Centrality strategy.

The impact of transmission rates (β) on the effectiveness of mosquito removal strategies was also significant. The simulations demonstrated that

higher transmission rates result in more widespread disease outbreaks, as indicated by the increased number of infected cities. This highlights the importance of timely and effective mosquito control measures, especially in regions experiencing higher temperatures and consequently higher mosquito activity.

Interestingly, the approach of combining multiple centrality measures into a single strategy performed well. This combined approach aimed to capture a broader range of influential cities and showed promising results, suggesting that integrating various centrality metrics can provide a comprehensive strategy for mosquito removal. Among the specific strategies, Combined Centrality was found to be the most effective, followed by Eigenvector Centrality. This indicates that focusing on the most connected cities and those connected to other influential cities provides significant disruption to the disease transmission network.

The removal of a small, manageable number of highly central cities (e.g., five cities) can significantly reduce the spread of diseases. This strategy is both practical and impactful, demonstrating that well-targeted interventions can yield substantial public health benefits without the need for widespread mosquito removal efforts.

Higher transmission rates, represented by higher β values in our simulations, correspond to conditions where mosquito activity is increased, often due to higher temperatures. This has significant implications for disease control, particularly in the context of climate change. Higher temperatures create more favourable conditions for mosquito breeding, potentially leading to larger mosquito populations. This can result in a higher base transmission rate (β), making it more challenging to control the spread of mosquito-borne diseases. The results indicate that with higher β values, the number of recovered cities is significantly higher, reflecting more widespread disease outbreaks. This suggests that in warmer climates, Europe could experience local epidemics of tropical diseases such as dengue fever, malaria, or Zika virus, which were previously considered as diseases imported from tropical regions.

The analysis shows that strategies based on centrality measures are crucial in controlling disease spread under high transmission rates. Degree centrality, in particular, appears to be the most effective strategy. This emphasises the need for targeted mosquito control efforts in highly connected cities to mitigate the risk of widespread epidemics. As climate change continues to raise global temperatures, public health strategies must adapt to the increased risk of tropical diseases spreading in non-tropical regions. This includes enhancing surveillance, improving mosquito control infrastructure, and developing adaptive strategies based on real-time data and centrality measures. The findings indicate that targeting cities with high degree centrality can significantly reduce disease transmission. However, the optimal strategy may vary based on specific network structures and transmission dynamics. Future research should explore adaptive strategies that dynamically adjust to changing conditions and incorporate real-time data for enhanced effectiveness.

4.2 Implications and Future Work

The findings from this study have several important implications for public health strategies and future research. Public health authorities should prioritise mosquito control efforts in cities identified with high Combined Centrality. These cities play a pivotal role in disease transmission networks, and their targeted removal can effectively reduce the spread of diseases like malaria, dengue fever, and Zika virus. This approach allows for efficient use of resources and maximises the impact of interventions. As global temperatures rise due to climate change, regions previously unaffected by mosquito-borne diseases may become susceptible. Public health strategies need to adapt to this changing landscape by enhancing surveillance and mosquito control infrastructure in both tropical and non-tropical regions. Proactive measures will be essential to prevent the spread of diseases in newly vulnerable areas.

Future research should focus on integrating real-time data into disease control models to develop adaptive strategies. Utilising real-time temperature, mosquito population, and disease incidence data can enable more responsive and effective interventions, tailored to current conditions and emerging trends. Additionally, investigating other factors that influence disease spread, such as human mobility patterns and local environmental conditions, could provide deeper insights and improve the effectiveness of control strategies.

The simulations in this report are based on certain assumptions and parameter choices, which may limit the generalisability of the findings. Future studies should seek to validate these results with empirical data from real-world mosquito-borne disease

outbreaks. Incorporating empirical data will enhance the robustness of the models and ensure that the recommendations are grounded in observed phenomena. As the dynamics of mosquito-borne disease transmission are influenced by various factors, including climate change, public health strategies must be adaptive and flexible. Developing strategies that can dynamically adjust based on real-time data and evolving conditions will be crucial for effective disease control. This includes continuous monitoring, rapid response to outbreaks, and adjusting control measures as needed.

In conclusion, the targeted removal of mosquitoes from cities with high centrality, particularly those with high Combined Centrality, is a highly effective strategy for controlling the spread of mosquito-borne diseases. As climate change continues to impact global temperatures and mosquito activity, it is essential to develop adaptive, data-driven public health strategies to protect populations and reduce disease transmission. The findings of this report provide a strong foundation for future research and practical applications in the field of mosquito-borne disease control.

References

[1] Smith, D.L., et al. (2012). "Mosquito population dynamics and the epidemiology of vector-borne diseases." *Nature*, 491(7423), 431-435.

[2] World Health Organization. (2016). *Dengue: guidelines for diagnosis, treatment, prevention, and control*. WHO.

A Removal Strategy

Algorithm 1 Centrality Removal Strategy

Require: Graph G, Number of cities to remove num_to_remove

Calculate degree centrality $C_D(v)$ for each node v in G
Calculate betweenness centrality $C_B(v)$ for each node v in G
Calculate closeness centrality $C_C(v)$ for each node v in G
Calculate eigenvector centrality $C_E(v)$ for each node v in G
Normalise each centrality measure:
for each node v in G **do**
    $C_D(v) \leftarrow \frac{C_D(v)}{\max(C_D)}$
    $C_B(v) \leftarrow \frac{C_B(v)}{\max(C_B)}$
    $C_C(v) \leftarrow \frac{C_C(v)}{\max(C_C)}$
    $C_E(v) \leftarrow \frac{C_E(v)}{\max(C_E)}$
end for
Calculate combined centrality score $C_{combined}(v)$ for each node v:

$$C_{combined}(v) = \frac{C_D(v) + C_B(v) + C_C(v) + C_E(v)}{4}$$

Sort cities by combined centrality score in descending order
Select the top num_to_remove cities based on the combined centrality score
for each city in the top num_to_remove **do**
    if city exists in G **then**
        Remove city from G
        Output: "Removed city: " + city
    else
        Output: "City " + city + " not found in the graph and cannot be removed."
    end if
end for
Output: "Cities removed for mosquito control."